AI-Powered Automated Proctoring System

sycet-004e	AI-Powered Automated Proctoring System	Python, Computer Vision	Build a webcam-based monitoring system for online exams. The software should use pre-trained models to detect multiple faces, head pose (looking away), and the absence of the candidate, logging suspicious events to a dashboard.

This project is an AI-based automated online exam monitoring system developed using Python and Computer Vision. The system detects suspicious activities during online examinations such as: Multiple faces in front of webcam Candidate looking away from screen Candidate absence Mobile phone detection Face mismatch All suspicious events are logged and displayed on an admin dashboard.

1. Problem Statement

Online exams face major challenges like cheating, impersonation, and lack of monitoring. Manual invigilation is difficult for large-scale exams. This system automates monitoring using AI and Computer Vision.

2. Objectives

Detect cheating activities automatically Monitor candidate through webcam Generate alerts for suspicious behavior Maintain exam integrity Store logs and evidence for review

3. Technologies Used

Programming Language: Python
Libraries: OpenCV, MediaPipe, face_recognition, NumPy, Flask
Database: SQLite / MySQL
Frontend: HTML, CSS, JavaScript
AI Techniques: Face Detection, Eye Tracking, Object Detection

4. System Architecture

Webcam → AI Processing → Suspicious Activity Detection → Event Logging → Dashboard Monitoring

5. Features

Feature	Description
Face Detection	Detects candidate face using webcam
Multiple Face Detection	Alerts when more than one face appears
Head Pose Detection	Checks if user is looking away
Absence Detection	Detects when candidate leaves
Dashboard	Shows logs and alerts

Screenshot Capture	Captures suspicious moments

6. Project Folder Structure

```
AI-Proctoring-System/
■
■■■ app.py
■■■ requirements.txt
■■■ templates/
■   ■■■ index.html
■■■ static/
■■■ logs/
■■■ screenshots/
■■■ models/
```

7. Installation Steps

```
1. Install Python 3.10+
2. Install required libraries:
   pip install opencv-python mediapipe flask face_recognition numpy
3. Run the application:
   python app.py
```

8. Sample Python Code

```
import cv2

camera = cv2.VideoCapture(0)

face_cascade = cv2.CascadeClassifier(
    cv2.data.haarcascades + 'haarcascade_frontalface_default.xml'
)

while True:
    ret, frame = camera.read()
    gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)

    faces = face_cascade.detectMultiScale(gray, 1.3, 5)

    if len(faces) > 1:
        print("Multiple faces detected!")

    for (x, y, w, h) in faces:
        cv2.rectangle(frame, (x,y), (x+w,y+h), (255,0,0), 2)

    cv2.imshow("AI Proctoring System", frame)

    if cv2.waitKey(1) & 0xFF == ord('q'):
        break

camera.release()
cv2.destroyAllWindows()
```

9. Advanced Features You Can Add

Voice detection during exams AI-based emotion analysis Mobile phone detection using YOLO Cloud dashboard Automatic report generation Live admin monitoring panel

10. Applications

Online university examinations Recruitment tests Certification exams Remote learning platforms

11. Advantages

Reduces cheating Saves manpower Real-time monitoring Accurate suspicious activity detection Improves exam security

12. Future Scope

Future improvements can include cloud deployment, AI analytics, behavior prediction, and integration with LMS systems like Moodle or Google Classroom.

Prepared for Hackathon / Academic Project